

Dynamics - **Technical Project**
Project 3. – “Friction Coefficient”

Objective:

A series of technical projects provide students with a hands-on experience to understand fundamental concepts in Dynamics through “learning by doing.” Throughout the ME 2330 course lectures, students will learn about basic principles, concepts, analytical approaches, and problem-solving skills in Dynamics. To supplement theories and “learning by lectures and problem-solving,” this project will help students connect the concepts and theories in class with experiments and “doing” as a group.

Summary:

This project is to measure your kinetic friction coefficient, μ_k . A group of 3~4 students comes up with their strategies to find a μ_k . A typical procedure could start with placing an object that you know its weight on a flat surface, then slightly and gradually change the surface angle until the object is at rest and starts sliding down the slope. Then, you measure the angle of the surface. You can find related friction coefficients with the object's known weight and the measured angle.

$$\bar{f} = \mu_k \cdot N$$

Where N is the normal force.

You may use two materials for the surface and your object choice. Then, you can say the friction coefficient of A material on the surface of B material.

Submission Guideline:

1. Write (type) a one or two-page document.
2. Record a short (less than 30 seconds) YouTube video.
3. The video shows the object in motion along the inclined surface.
4. No cover page is necessary.
5. Take a group photo.
6. Submit 1) a document, 2) a video link, and 3) a group photo through iLearn.

Measurement Guideline:

1. Your group can pick any non-rolling objects and materials.
2. At least 2 or 3 experiments are recommended for accuracy.
3. Work as a group. (Each individual plays a specific role)
4. Always be careful! (Safety is the first)
5. Brainstorm and discuss first before you go out for an experiment.